

Hannah Potter

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Research Interests

My primary research interest is in improving programmer comprehension of code, program analyses, and programming languages, with my most recent focus being on comprehension of program changes (e.g., pull requests).

Research Projects

Understanding Code Changes

2025 – present

Exploring ways to support programmer comprehension of code changes (e.g., pull requests).

- Investigating using *change tours* - an interleaving of narratives with related groups of change sets - to guide a developer through understanding a set of changes.
- Interested in exploring applications for human-written and LLM-generated code.

Infrastructure for Mining Real-World Bugs

2025 – present

Improving the [Defects4J](#) [🔗](#) bug-mining infrastructure.

- Utilize a project's original build system to reduce loss of bugs due to build issues.
- Support more rapid and automatic resolution of build issues.
- Migrate from *ant* to native *java* and *javac* build and execution targets of mined projects to support more customized set-ups for software engineering researchers using Defects4J.

Mutant Explanations

2023 – 2025

Added visualizations to support understanding static and dynamic information about undetected mutants and tests to detect these mutants.

- Classroom study found that visualizations helped students perform more effective mutation testing, with lower-performing students seeing a boost in particular.

Tools to Support Experienced Programmers Learning a New Programming Language

2020 – 2022

Implemented/enhanced 3 programming tools in [Hazel](#) [🔗](#), a live functional programming environment with holes.

- The Cursor Inspector displays typing information about the expression or pattern at the cursor.
- The Strategy Guide is designed to guide novices to functional programming through a type-driven development strategy.
- ExplainThis is a contextualized programming language documentation tool, which provides information about the syntactic form, an explanation, and examples for the term at the cursor.

Papers

Evaluating the Impact of Scaffolding and Visualizations for Mutation Testing Exercises in Software Engineering Education

2025

Hannah Potter, Ana C.R. Paiva, Domenico Amalfitano, Anna Rita Fasolino, Porfirio Tramontana, René Just [10.1145/3696630.3727240](#) [🔗](#) (Proceedings of the ACM International Conference on the Foundations of Software Engineering (Software Engineering Education @ FSE))

Equivalent Mutants in the Wild: Identifying and Efficiently Suppressing Equivalent Mutants for Java Programs

2024

Benjamin Kushigian, Samuel J. Kaufman, Ryan Featherman, **Hannah Potter**, Ardi Madadi, René Just [10.1145/3650212.3680310](#) [🔗](#) (Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA))

Live Pattern Matching with Typed Holes

2023

Yongwei Yuan, Scott Guest, Eric Griffis, **Hannah Potter**, David Moon, Cyrus Omar

[10.1145/3586048](#) [🔗](#) (Proceedings of the ACM on Programming Languages (OOPSLA @ SPLASH))

Contextualized Programming Language Documentation

2022

Hannah Potter, Ardi Madadi, René Just, Cyrus Omar

[10.1145/3563835.3567654](#) [🔗](#) (ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward! @ SPLASH))

Hazel Tutor: Guiding Novices Through Type-Driven Development Strategies

2020

Hannah Potter, Cyrus Omar

hazel.org/hazeltutor-hatra2020.pdf [🔗](#) (Human Aspects of Types and Reasoning Assistants (HATRA @ SPLASH))

Industry Experience

Lucid Software, Software Engineer Intern

Greater Salt Lake City, UT

- Worked on the LucidChart Core Experience team working to make the main experience in LucidChart smooth and powerful.

May 2019 – Aug 2019

Lucid Software, Software Engineer Intern

Greater Salt Lake City, UT

- Worked on the Data Service development team to develop tools and services to make data a more integral part of the LucidChart experience.

May 2018 – Aug 2018

Lucid Software, Software Engineer Intern

Greater Salt Lake City, UT

- Worked on the LucidPress Core Experience development team improving core features of the product.

May 2017 – Aug 2017

Lucid Software, Software Engineer Intern

Greater Salt Lake City, UT

- Worked on the Automated Testing team and on an internal tools project.

May 2016 – Aug 2016

Awards

- 2nd Place in ECOOP/ISSTA Student Research Competition 2023
- Best Paper Award @ OOPSLA 2023
- NSF GRFP Honorable Mention 2021
- 3rd Place Team for Senior Capstone Project 2019
- 3rd Place Team in Lucid Software Employee Hackathon 2017
- National Merit Scholar 2015
- University of Utah President's Scholarship 2015

Education

PhD University of Washington, Computer Science and Engineering

Seattle, WA

- GPA: 3.91/4.0
- MS in Computer Science and Engineering (2023)
- Anticipated graduation 06/27

2021 – present

MSE University of Michigan, Computer Science and Engineering

Ann Arbor, MI

- GPA: 4.0/4.0

2019 – 2021

BS University of Utah, Computer Science

Salt Lake City, UT

- Magna Cum Laude
- Dean's List (All Semesters)
- Honors BS
- Major GPA: 4.0/4.0, Cumulative GPA: 3.982/4.0

2015 – 2019

Teaching Experience

University of Washington, CSE Teaching Assistant

2021 – present

Led discussions, conducted help hours and test reviews for students, and graded assignments and tests. Additionally assisted in the development and refinement of course exercises.

- Software Engineering (CSE 403) - Autumn 2025
- Building Data Analysis Pipelines (CSEP 590) - Autumn 2024
- Advanced Topics in Software Systems (CSEP 504) - Autumn 2022
- Software Engineering (CSE 403) - Spring 2022
- Software Design and Implementation (CSE 331) - Winter 2022
- Compiler Construction (CSEP 501) - Autumn 2021

University of Michigan, EECS Graduate Student Instructor

2020 – 2021

Developed assignment and discussion course materials. Led discussions, conducted help hours for students, and graded assignments and tests.

- Advanced Programming Languages (EECS 590) - Winter 2021
- Programming Languages (EECS 490) - Fall 2020
- Programming Languages (EECS 490) - Winter 2020

University of Michigan, Engineering Teaching Consultant

2020 – 2021

Consulted with student instructors about good teaching practices and conducted midterm student feedback reviews. Led a cohort of new student instructors through discussions of good teaching practices, including inclusive teaching.

University of Utah, CS Teaching Assistant

2017 – 2019

Led labs, conducted help hours for students, conducted test reviews, and graded assignments and tests.

- Compilers (CS 4470) - Spring 2019
- Software Practice (CS 3500) - Fall 2018
- Introduction to Object-Oriented Programming (CS 1410) - Spring 2018
- Software Practice (CS 3500) - Fall 2017

Service and Outreach Experience

PhD K-12 Outreach Coordinator, University of Washington

2022 – 2025

- Led efforts to organize UW CSE PhD student outreach efforts to K-12 students in the greater Seattle area. Included organizing and hosting regular organizational meetings and workshops.
- Co-led efforts to create a centralized Allen School High School Summer Internship Program for Summer 2025, including organizing and hosting regular meetings for labs participating in the program to create applications, rubrics, and procedures for application review.
- Organized researcher involvement in UW CSE participation in CS Ed Week and Admitted Student Day, as well as lab activities for field trips to the Allen School.

Undergraduate Student Advisory Committee for the School of Computing, University of Utah

2016 – 2019

- Planned events, served as representatives for students to faculty/administration, participated in retention-promotion-tenure reviews of professors.
- Specifically focused on breaking down barriers between professors and students and building an inclusive and supportive environment for all CS undergraduate students.
- Served as Chair for the 2018-2019 academic year.

Student Volunteer, Conferences

- FSE/ISSTA 2025
- ECOOP/ISSTA 2023
- SPLASH 2022